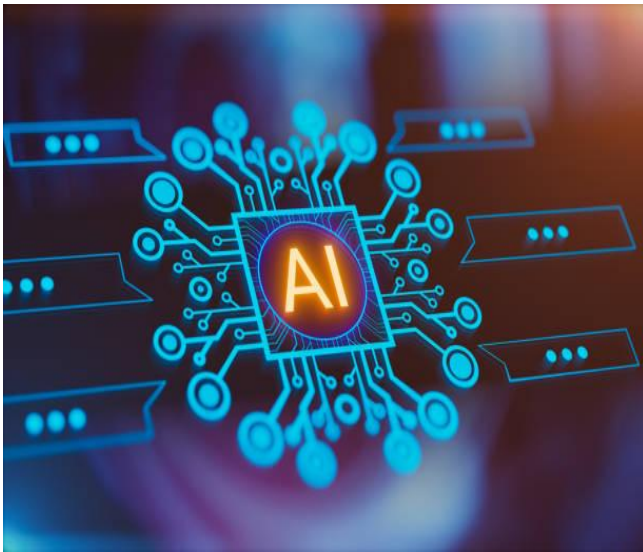




CANCER TYPES AND SYMPTOMS DATA

A Structured Reference for Clinical Decision
Support Systems

ABSTRACT

This document provides an explanation on what data we used to build the model, a structured and comprehensive list of cancer types and their associated symptoms, categorized by gender. It serves as a reference resource for clinical decision support and system design purposes.

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Table of Contents

Cancer types	2
The cancer types list (samples from NCR-2023).....	3
Female	3
Male	3
Both genders.....	3
Symptom lists for each cancer types	4
Symptoms for female:.....	4
Breast Cancer.....	4
Cervical Cancer	5
Uterine Cancer	6
Vulvar Cancer	6
Symptoms for male:	7
Prostate Cancer	7
Bladder Cancer.....	7
Symptoms for both genders:	8
Basal Cell Carcinoma (BCC)	8
Squamous Cell Carcinoma (SCC)	8
Melanoma	9
Colorectal Cancer.....	9
Primary Site Unknown (CUP).....	10
Non-Hodgkin Lymphoma.....	10
Lung Cancer	11
Oesophageal Cancer.....	12

Cancer types

We did research on the internet of finding a list of all **cancer types** diagnosed in South Africa latest and we came across the National-Cancer-Registry-2023 which was the latest available. This was very helpful and trusted. This registry contains so much data, but what was the most interesting to us was the top diagnosed cancer type. This data was going to be our building block for developing the model. What we did we took top 10 from both genders and research about their symptoms.

Symptoms are the most important part of our system therefore we had to take cancer types of symptoms from the most trusted sources therefore for our cancer types we took their symptoms from Wikipedia, WHO, and many more government trusted sites. So basically, we have a cancer type (e.g., Breast Cancer, Lung Cancer) and each Cancer type has its symptoms that belong to it. In terms of relationships a cancer type can have many symptoms, but a symptom can belong to many cancer types.

The data below is the data used to train and develop the model, cancer types and symptoms. Cancer types are retrieved from the [National-Cancer-Registry-2023](#), and the symptoms of those cancer types are retrieved from the Wikipedia.

The cancer types list (samples from NCR-2023)

Female

1. Breast
2. Cervix
3. Uterus
4. Vulva

Male

1. Prostate
2. Bladder

Both genders

1. Basal Cell Carcinoma (BCC)
2. Squamous Cell Carcinoma (SCC) of skin
3. Colorectal
4. Primary site unknown
5. Melanoma
6. Non-Hodgkin lymphoma
7. Lung
8. Oesophagus

Symptom lists for each cancer types

Symptoms for female:

Breast Cancer

- Lump in breast
- Breast swelling
- Breast pain
- Skin dimpling
- Skin thickening
- Skin redness
- Skin dryness
- Nipple pain
- Nipple inversion
- Nipple discharge
- Swelling of lymph nodes (underarm, collarbone)
- Inflammatory swelling and redness of breast
- Nipple/areola irritation (red, scaly)
- Bone pain (if spread)
- Bone fractures
- Abdominal pain (if spread to liver)
- Nausea/vomiting (liver/brain spread)
- Jaundice (yellow skin)
- Chest pain (lung spread)
- Shortness of breath
- Persistent cough
- Headache (brain spread)
- Seizures
- Vision problems

- Speech problems
- Memory changes
- Behavior changes

Cervical Cancer

- No early symptoms (often)
- Vaginal bleeding
- Bleeding after sex
- Vaginal mass
- Pain during sex
- Vaginal discharge
- Bleeding after douching
- Bleeding after pelvic exam
- Loss of appetite
- Weight loss
- Fatigue
- Pelvic pain
- Back pain
- Leg pain
- Swollen legs
- Heavy vaginal bleeding
- Bone fractures
- Urine leakage from vagina
- Feces leakage from vagina
- Flank pain (kidney blockage)
- Blood clots in legs
- Rectal bleeding
- Blood in urine

Uterine Cancer

- Abnormal vaginal bleeding
- Abnormal vaginal discharge
- Heavy menstrual bleeding
- Bleeding between periods
- Bleeding after menopause
- Pelvic pain
- Pelvic pressure

Vulvar Cancer

- Itching on vulva
- Burning sensation on vulva
- Vulvar bleeding
- Vulvar skin color changes (redder or whiter)
- Vulvar rash
- Vulvar warts
- Vulvar sores
- Vulvar lumps
- Vulvar ulcers
- Vulvar discharge
- Pelvic pain
- Pain during urination
- Pain during sex
- Lump on labia majora
- Irritation of vulva
- Bartholin gland painful lump

Symptoms for male:

Prostate Cancer

- Early stage: no symptoms
- Erectile dysfunction
- Blood in urine
- Blood in semen
- Trouble urinating
- Frequent urination
- Weak urine stream / slow flow
- Advanced: fatigue
- Unexplained weight loss
- Back pain
- Bone pain (pelvis, hips, ribs, spine, head, neck)
- Bone fracture (from metastasis)
- Leg weakness
- Foot weakness
- Paralysis (from spinal cord compression)

Bladder Cancer

- Visible blood in urine (painless)
- Microscopic blood in urine (only in tests)
- Pain while urinating
- No symptoms (detected by scan)
- Blocked urine flow → swollen kidneys, flank pain
- Bone metastasis: bone pain, fractures
- Lung metastasis: cough, coughing blood, breathlessness, infections
- Liver metastasis: abdominal pain, swelling, jaundice, itching, weight loss
- Swollen lymph nodes in abdomen/groin → pain, swelling

Symptoms for both genders:

Basal Cell Carcinoma (BCC)

- Shiny pearly skin nodule
- Red skin patch
- Skin thickening
- Scar-like skin changes
- Raised smooth pearly bump
- Pearly skin nodule
- Visible blood vessels in bump
- Crusting in tumor center
- Bleeding in tumor center
- Sore that does not heal

Squamous Cell Carcinoma (SCC)

- Skin changes not healing
- Skin ulcer
- Discolored skin
- Mole changes (jagged edges, bigger size, color change, bleeding, texture change)
- Painful lesion
- Itchy lesion
- Burning lesion
- Large brown spot with darker speckles
- Red scaly patch on skin
- Thickened skin patch
- Firm hard nodule
- Dome-shaped nodule (keratoacanthoma-like)
- Ulcer on skin
- Bleeding skin lesion
- Large growing mass (if untreated)

Melanoma

- Multicolored mole (brown to black)
- Pink/red/fleshy mole (amelanotic)
- Mole changes (size, shape, color, elevation)
- New mole in adulthood
- Pain at mole site
- Itchy mole
- Ulcerated mole
- Redness around mole
- Bleeding mole
- Asymmetrical mole
- Irregular borders
- Large diameter mole (>6mm)
- Mole evolving over time
- Elevated lump
- Firm lump
- Growing lump
- Loss of appetite (if spread)
- Nausea (if spread)
- Vomiting (if spread)
- Fatigue
- Brain metastases (headache, seizures, confusion)
- Liver/intestine/bone/lung/lymph node metastases

Colorectal Cancer

- Constipation (worsening)
- Blood in stool
- Thin stool (narrow caliber)

- Loss of appetite
- Weight loss
- Nausea
- Vomiting
- Rectal bleeding
- Anemia
- Bowel habit change

Primary Site Unknown (CUP)

- Masses or lumps in body
- Swelling in body
- Rapid tumor growth
- Unusual metastasis sites
- Enlarged lymph nodes (armpit, chest, abdomen, pelvis)

Non-Hodgkin Lymphoma

- Enlarged lymph nodes
- Lumps under skin
- Fever
- Night sweats
- Weight loss
- Fatigue
- Bone pain
- Chest pain
- Itchy skin
- Skin lumps (itchy, red, purple)
- Weakness (if in brain)
- Seizures

- Thinking problems
- Personality changes

Lung Cancer

- Early: no symptoms
- Persistent cough (new or worsening)
- Coughing blood (streaks or large amounts)
- Shortness of breath
- Chest pain (dull, persistent)
- Loss of appetite
- Weight loss
- Fatigue
- Fever
- Night sweats
- Difficulty swallowing
- Hoarseness (voice change)
- Shoulder pain (radiating down arm – Pancoast tumor)
- Swollen lymph nodes (above collarbone)
- Swelling of face/upper body (superior vena cava syndrome)
- Fluid around heart (arrhythmia, heart failure)
- Brain metastasis: headache, nausea, vomiting, seizures, confusion
- Bone metastasis: bone pain, fractures, spinal cord compression
- Liver metastasis: enlarged liver, abdominal pain, fever, jaundice
- Hormone syndromes:
 - Hypercalcemia → nausea, constipation, thirst, confusion
 - Hyponatremia (low sodium)
 - Hypokalemia (low potassium)
- Nail clubbing

- Joint pain, bone thickening (hypertrophic osteoarthropathy)
- Blood clots (migratory thrombophlebitis, heart clots, DIC)
- Autoimmune effects: muscle weakness, neuropathy, brain inflammation

Oesophageal Cancer

- Difficulty swallowing (solids → liquids)
- Pain when swallowing
- Weight loss
- Loss of appetite
- Pain behind breastbone
- Severe heartburn-like pain
- Hoarse/raspy cough
- Regurgitation of food
- Nausea
- Vomiting
- Coughing when eating/drinking
- Vomiting blood
- Aspiration pneumonia
- Airway obstruction
- Superior vena cava syndrome
- Hypercalcemia (high calcium)
- Metastasis:
 - Liver → jaundice, abdominal swelling
 - Lungs → breathing difficulty, pleural effusion
 - Bone → pain, fractures